

GEOA-Game_Project25-26

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In this project I've been assigned to use 3D PPGA operations and transformations to create the start of a third person 3D game with movement, collisions and interesting camera. As my extra feature I've decided to use a projectile, that the player can use and throw at different angles. This sphere can interact with their environment and score point by hitting the target.

To highlight certain parts, I've extracted 3 formulas that I used in the project to create this result.

The Sandwich Product for Point Transformation

The Formula: $\text{nextPos} = (T * \text{currentCenter} * \sim T).Grade3();$

I used this to transform the sphere center (a trivector) based on the translation Motor. By extracting the third grade, I can assign it again to the sphere.

The Wedge Product for Distance Calculation

The Formula: $\text{distance} = (m_Planes[index] \wedge \text{currentPos}).e0123();^1$

This formula was used multiple times to calculate the distance between 2 objects. In this project it was used for the collision's calculations.

Translation via Line Duality

The Formula: $\text{Motor } T = \text{Motor}::\text{Translation}(\text{bulletSpeed} * \text{deltaTime}, -!\text{bulletDirection});$

I used this to create the path my sphere had to follow.

¹ Only works if it's normalized